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Assignment
**Plant Disease Localization and
Classification on Field Images**

**Programming Foundation for Data Analytics and
Visualization**

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1 Introduction

Automatic recognition of plant disease from photographs is usually presented as an image classification problem, but field photographs may contain several overlapping plants, soil, shadows, and visually similar background vegetation. The task must therefore identify both the region being described and its disease or health state.

This report studies the problem in its two-level form. Given a field photograph, the system must first *localize* each annotated plant region and then *classify* its disease or health state. The two stages are evaluated separately because a missed region never reaches the classifier, whereas a correctly localized region may still be mislabelled.

1.1 Objectives

The objectives are to characterize FieldPlant, build a classical colour-and-morphology baseline, train the Ultralytics YOLO26n detector [3] on the same split, and compare both tracks under one matching protocol.

1.2 Contributions

The report quantifies class imbalance and the limits of colour-based localization, evaluates six proposal configurations, two feature representations, and ten classifier candidates, and compares the selected classical pipeline with class-aware YOLO26n. Its attribution analysis shows that localization coverage explains most of the performance gap.

2 Dataset and Experimental Protocol

2.1 Source and acquisition

All experiments use FieldPlant [1], a public collection of plantation photographs with plant-region bounding boxes, released through Roboflow Universe [2]. The study uses a YOLO-format export of that release.

The export used here contains **5,156 images** and **8,580 annotated boxes** distributed over **27 classes**: five cassava, sixteen corn, and six tomato states. Because the export places every image in one directory, the notebook constructs the validation and test partitions.

2.2 Image-level stratified split

Images, rather than individual boxes, are split so that all annotations from one photograph stay together. Majority-class stratification produces a **70 / 15 / 15** train, validation, and test split. Corn Charcoal and Tomato bacterial wilt are pinned to training because their support is too small for stratification; Table 1 gives the outcome.

Partition	Images	Boxes	Role
Train	3,609	5,997	Fits every model, scaler, PCA basis, and sampling tier
Validation	773	1,318	Selects proposal method, features, and classifier
Test	774	1,265	Evaluated once, for the selected configuration only
Total	5,156	8,580	

Table 1: Image-level 70 / 15 / 15 split actually materialized by the notebook. Boxes follow their parent image file, so the box proportions deviate slightly from 70 / 15 / 15.

2.3 Integrity audit

An independent parser verifies all **5,156 images** and **8,580 annotation rows**. No image fails to decode, and no annotation has an unknown class, invalid coordinate, or degenerate box.

2.4 Evaluation discipline

The refreshed exploratory analysis and earlier modelling run use the same materialized split, verified by content signatures, but were not executed in one continuous run.

Four rules are enforced throughout and are the reason several quantities in this report are lower than they would otherwise appear.

1. **Train-only fitting.** Every quantity estimated from data is estimated on the training partition alone. This includes the **StandardScaler** used by the linear classifiers, the PCA basis used in Section 3.7, the class-frequency counts that produce the balanced class weights, and the oversampling tiers of Section 4.4. Validation and test features are transformed by these fitted objects, never used to fit them.
2. **Validation selects, test confirms.** The proposal configuration, the feature representation, and the classifier hyperparameters are chosen using validation results only. The test partition is evaluated once, for the selected configuration.
3. **No augmentation or oversampling outside training.** Augmentation transforms and repeated-path manifests apply exclusively to training images. Validation and test directories are never modified. The notebook verifies this by comparing content signatures of those directories before and after training.
4. **Fixed seeds.** A single seed, 42, governs the split, the class-capped feature sampling, the classifier initializations, and the detector training run, and the detector is trained in deterministic mode.

3 Exploratory Data Analysis

SHA-256 comparison finds **113 groups of byte-identical files spanning partitions**. This leakage makes held-out results optimistic by an unknown amount; the split is left unchanged for reproducibility, and Section 7 records the consequence. Hashing detects exact duplicates only, not recompressed or near-duplicate photographs.

3.1 Class imbalance and evaluation coverage

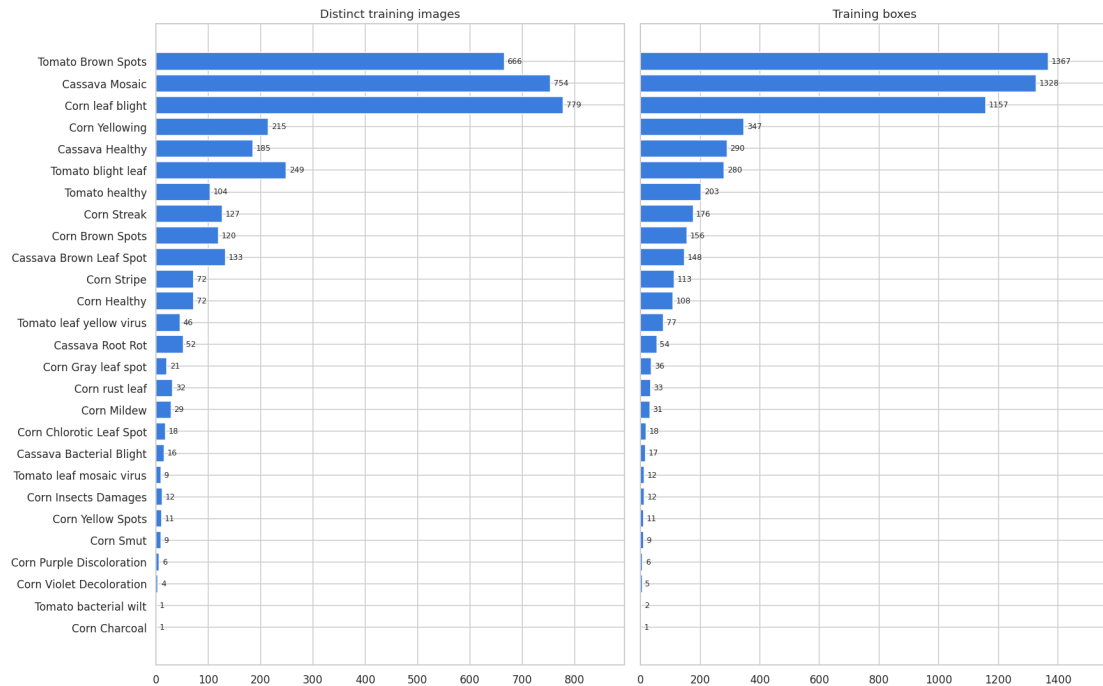


Figure 1: Distinct training images (left) and training boxes (right) for each of the 27 classes, sorted by box count. The two panels share a vertical axis, so the horizontal difference between them shows how many boxes a class contributes per photograph.

Figure 1 shows a distribution that is not merely skewed but skewed across three orders of magnitude. The three largest classes, Tomato Brown Spots with 1,367 training boxes, Cassava Mosaic with 1,328, and Corn leaf blight with 1,157, together hold 3,852 of the 5,997 training boxes, that is **64.2%** of the training annotations. At the other end, **ten of the 27 classes have fewer than 20 training boxes**, and the smallest, Corn Charcoal, has a single box in a single photograph.

The imbalance also determines what can be measured. Table 2 gives the per-class box counts in each partition. Two classes, **Corn Charcoal** and **Tomato bacterial wilt**, have no boxes at all in validation or test. For these two classes the report can describe training behaviour but cannot report held-out performance of any kind, and any per-class metric printed for them is an artefact of the averaging convention rather than a measurement. Several further classes, such as Corn Purple Discoloration and Corn Violet Decoloration with one validation box each, are technically evaluable but their per-class scores are determined by a single detection.

Class	Tr	Va	Te	Class	Tr	Va	Te
Cassava Bacterial Blight	17	5	3	Corn Streak	176	35	41
Cassava Brown Leaf Spot	148	34	29	Corn Stripe	113	25	24
Cassava Healthy	290	70	57	Corn Violet Decoloration	5	1	1
Cassava Mosaic	1,328	285	284	Corn Yellow Spots	11	3	2
Cassava Root Rot	54	11	12	Corn Yellowing	347	75	73
Corn Brown Spots	156	31	33	Corn leaf blight	1,157	256	239
Corn Charcoal	1	0	0	Corn rust leaf	33	7	9
Corn Chlorotic Leaf Spot	18	4	4	Tomato Brown Spots	1,367	298	302
Corn Gray leaf spot	36	10	8	Tomato bacterial wilt	2	0	0
Corn Healthy	108	25	19	Tomato blight leaf	280	62	62
Corn Insects Damages	12	3	2	Tomato healthy	203	43	33
Corn Mildew	31	6	6	Tomato leaf mosaic virus	12	4	3
Corn Purple Discoloration	6	1	1	Tomato leaf yellow virus	77	22	16
Corn Smut	9	2	2	Total	5,997	1,318	1,265

Table 2: Annotated boxes per class and partition. Classes are listed in the order defined by the dataset. The two rows with zero validation and test boxes cannot be evaluated on held-out data.

Because boxes from one photograph are correlated and the three largest classes dominate the counts, the report uses macro- F_1 with per-class support and reweights or resamples training.

3.2 Objects per image

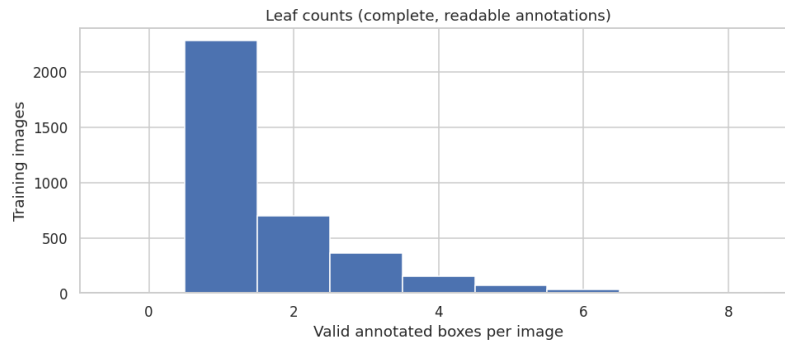


Figure 2: Number of valid annotated boxes per training photograph, over the 3,609 images with complete and readable annotations.

Figure 2 answers a question that constrains the classical pipeline directly. Across the 3,609 eligible training images, **63.3% contain exactly one annotated object**, **36.7% contain more than one**, and **17.5% contain more than two**. No eligible image has zero boxes.

The classical proposal stage of Section 4.3 retains at most two boxes per image. Using Equation (14) with $k = 2$ and assuming perfect box placement, it could match at most **82.3%** of the training annotations: images with three or more objects necessarily lose the remainder. This count-only bound isolates the retention rule from mask quality; detectors without a two-box limit retain a structural advantage on multi-object images.

3.3 Box geometry and scale

Detector and classifier input sizes are only meaningful relative to the size of the objects. Table 3 summarizes the geometry of the 5,997 training boxes in three forms: their size in original

pixels, their physical aspect ratio, and their *nominal* size after an aspect-preserving resize of the photograph to a 640-pixel long side, which is the input convention used by the detector.

Quantity	Min	5%	Median	Mean	95%	Max
Box width (px)	78.2	525.2	1,476.0	1,688.4	3,398.7	4,608.0
Box height (px)	57.5	519.0	1,580.0	1,875.3	4,158.0	4,608.0
Aspect ratio (width / height)	0.10	0.39	0.92	1.08	2.22	6.33
Nominal short side (px)	11.8	88.3	219.3	247.3	472.5	639.3

Table 3: Distribution of training box geometry. “Nominal short side” is the shorter box dimension after an aspect-preserving resize of the photograph to a 640-pixel long side.

The annotations are large. The median box occupies a substantial part of its photograph, and at the nominal 640-pixel working resolution the median box still has a short side of **219.3 pixels**. Only **0.02%** of boxes fall below a 16-pixel short side and **0.08%** below 32 pixels. Small-object detection is therefore not the binding constraint here. These nominal sizes ignore letterboxing, rounding, and augmentation, and boxes can overstate the extent of non-rectangular plant tissue.

Aspect ratio is a different matter. The central 90% of boxes span ratios from 0.39 to 2.22 and the extremes reach 0.10 and 6.33, meaning some annotations are roughly ten times taller than they are wide. Because the classical classifier resizes every crop to a fixed 128×128 square, such boxes are geometrically distorted before any feature is computed.

3.4 Spatial distribution of annotations

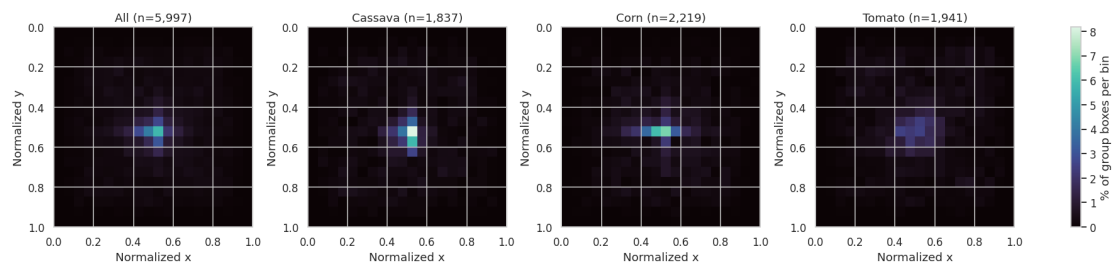


Figure 3: Normalized annotation centres over the training partition: all boxes, then cassava, corn, and tomato separately. Each cell reports the percentage of that group’s boxes, on a shared colour scale. The vertical axis increases downwards, matching image coordinates.

Figure 3 shows that annotation centres are strongly concentrated: **65.6% of training box centres fall inside the central quarter of the image area**, defined as the region where both normalized coordinates lie in $[0.25, 0.75]$. The concentration is sharpest for cassava and corn. Tomato centres are visibly more dispersed, which is consistent with tomato photographs containing several annotated regions more often.

This pattern is a statement about how the corpus was photographed and annotated. The photographer aimed at the plant of interest and the annotator labelled it, so the label lands near the centre. It does not follow that diseased plants occur near image centres in general, and a model that learned a centre prior would be learning a property of the acquisition process. Error inspection therefore includes off-centre and border-touching objects.

3.5 Classes that occur together

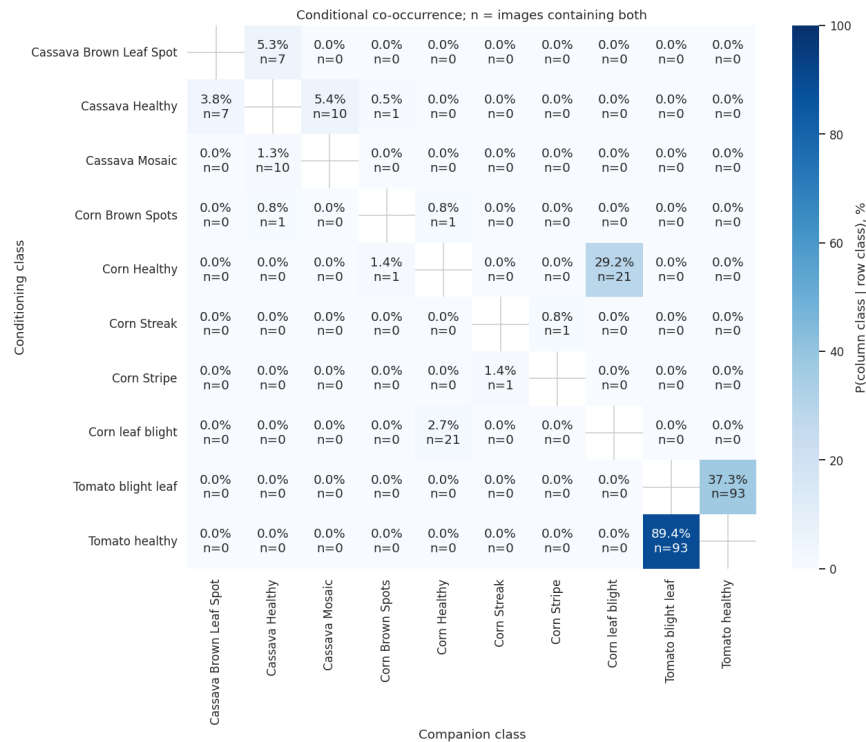


Figure 4: Conditional co-occurrence for class pairs found in the same training photograph. Cell (a, b) reports the percentage of training photographs of the row class that also contain the column class, together with the supporting image count. The matrix is asymmetric and the diagonal is masked.

Only **seven distinct class pairs** ever appear together in a training photograph, and Table 4 lists all of them. The relationship is dominated by a single pair: **93 photographs contain both Tomato healthy and Tomato blight leaf**, which is exactly the situation one expects when a diseased and a healthy leaf of the same plant appear in one frame.

Because Figure 4 is conditional, direction matters. Those same 93 images constitute **89.4% of the photographs containing Tomato healthy** but only **37.3% of those containing Tomato blight leaf**. Reading the matrix symmetrically would therefore misstate the relationship in one of the two directions. Three of the seven pairs rest on a single photograph each, and one of them, Cassava Healthy together with Corn Brown Spots, crosses crop boundaries and is worth inspecting as a possible annotation error rather than treated as a real biological co-occurrence.

Class A	Class B	Images
Tomato blight leaf	Tomato healthy	93
Corn Healthy	Corn leaf blight	21
Cassava Healthy	Cassava Mosaic	10
Cassava Brown Leaf Spot	Cassava Healthy	7
Cassava Healthy	Corn Brown Spots	1
Corn Brown Spots	Corn Healthy	1
Corn Streak	Corn Stripe	1

Table 4: All class pairs co-occurring in at least one training photograph.

Co-occurrence supported by one or two photographs is unstable and does not validate the annotations. It nevertheless confirms that splitting by image is mandatory; cross-crop pairs also warrant manual inspection.

3.6 Representative photographs and classifier inputs

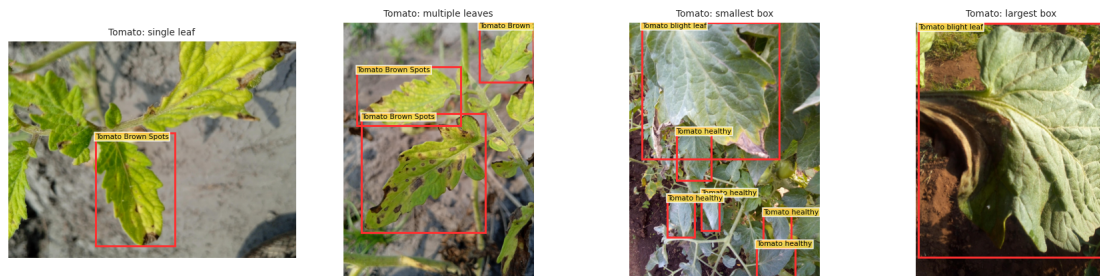


Figure 5: Four tomato training photographs with ground-truth boxes in red: one annotated leaf, several annotated leaves, the smallest tomato annotation, and the largest.

Figure 5 is included to prevent a misleading mental image of the corpus. A single representative photograph does not exist: the same class appears as one clean, well-separated leaf, as a cluster of overlapping leaves each carrying its own annotation, as a small local region inside a wider scene, and as an object filling almost the whole frame. Any localization heuristic tuned on the first case will not transfer to the second.

Inspecting one or two original crops from every class also surfaces a scope question. The crops displayed for **Corn Charcoal** and **Corn Smut** include ears and tassels rather than leaf blades alone, so the annotations are not restricted to leaves even though the task is frequently described in those terms. A montage cannot establish how often this happens, and no annotation was removed on the strength of it. The point is recorded so that the task definition used in Section 4 speaks of plant regions rather than leaves. One further record was found in which annotations of two different crops, cassava and corn, occur in the same photograph. These examples also motivate including crowded scenes in qualitative evaluation.

3.7 Structure of the handcrafted feature space

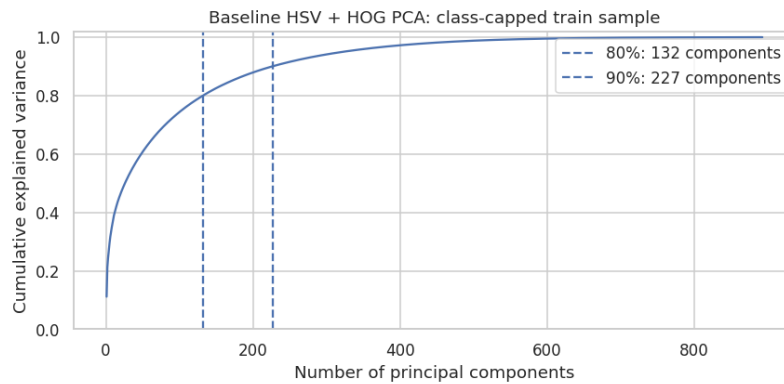


Figure 6: Cumulative explained variance of a PCA fitted to standardized HSV and HOG features of class-capped training crops.

To understand what the classical representation offers before any classifier is fitted, the baseline 1,812-dimensional HSV and HOG descriptor of Section 4.3.4 is computed on a class-capped training sample of at most 50 crops per class, yielding **893 usable crops** from 783 photographs, and a PCA is fitted to the standardized result.

Figure 6 shows that the representation is genuinely high dimensional. The first two principal components retain only **21.2%** of the variance. Reaching 80% requires **132 components** and reaching 90% requires **227**. The practical consequence is about interpretation rather than about modelling. A two-dimensional scatter plot of the first two components shows less than a quarter of the variation, so classes that overlap in that projection have not been shown to be inseparable. They have only been shown to be indistinguishable in a projection that discards four fifths of the signal.

PCA measures variance rather than class separation and depends on the standardized, class-capped sample. Low-dimensional projections are therefore used for inspection only; separability claims come from held-out classifier metrics.

3.8 Feasibility of colour-based localization

The classical track localizes objects by colour and morphology, so its feasibility should be probed before it is built in full. The probe applies the same Excess Green and HSV filtering pipeline and the same top-two retention rule used later, on 100 seeded training photographs containing 153 annotated objects, and matches proposals to annotations with the greedy procedure of Section 5.4.

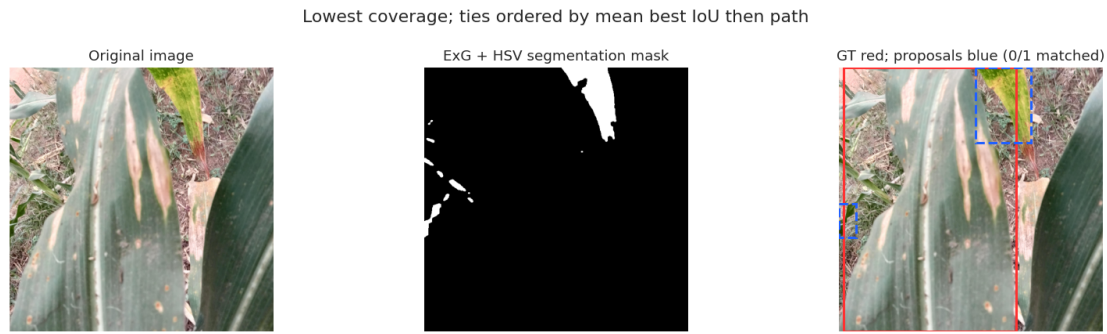


Figure 7: A low-coverage example. Left to right: the photograph with the red ground-truth box, the vegetation mask, and the retained proposals in blue. Discoloration and background foliage fragment the mask, and no retained proposal reaches an IoU of 0.5 with the annotation.

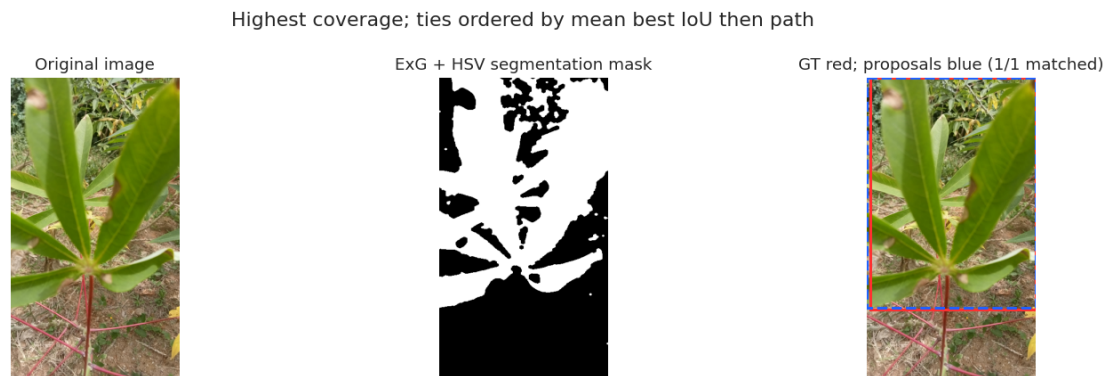


Figure 8: A high-coverage example. A clean green foreground against a contrasting background yields one proposal that follows the ground-truth box closely.

The two examples show the mechanism in both directions. In Figure 8 the object is a compact green region against a dissimilar background, the mask has one connected component, and the fitted box nearly coincides with the annotation. In Figure 7 the leaf is discoloured, which removes part of it from a green-based mask, while healthy background vegetation is retained. The mask splits into fragments and the two largest of them describe regions that the annotation does not.

Aggregated over the sample, the second behaviour is the common one. The procedure keeps 192 proposals, an average of 1.92 per photograph, and matches only **45 of the 153 annotated objects**, a coverage of **29.4%**, leaving **108 unmatched**. The mean best IoU per annotation, as defined in Equation (13), is **0.375**, comfortably below the 0.5 matching threshold. In other words the typical proposal overlaps its nearest annotation partially rather than either missing it entirely or capturing it.

This training-only probe is small, and mean best IoU is not one-to-one like coverage. It still identifies localization as the expected bottleneck, motivating separate classification measurements on annotation crops and proposal crops.

4 Proposed Method

4.1 Two-level architecture

The classical track proposes regions using colour and morphology and then classifies their crops; YOLO26n predicts boxes and classes jointly. The explicit separation permits stage-wise attribution, while the shared matching stage in Figure 9 makes the final outputs comparable.

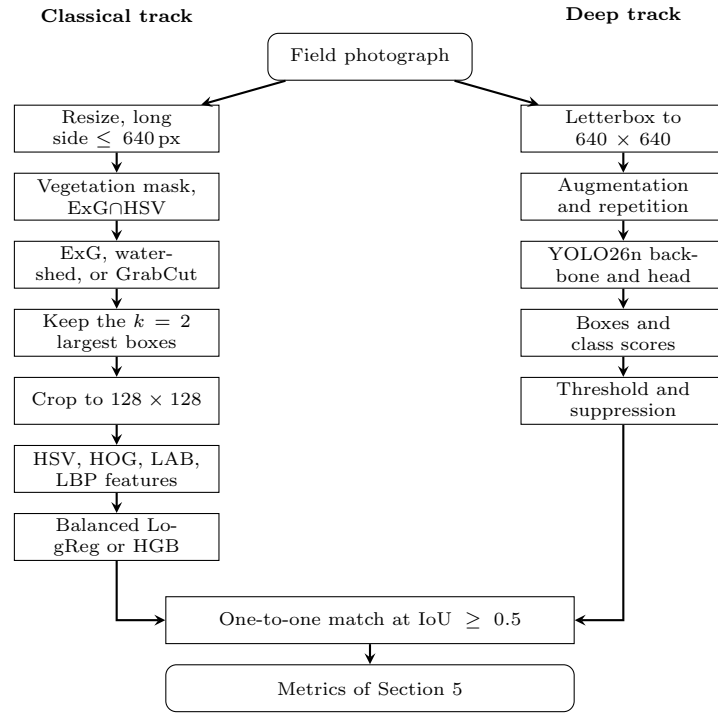


Figure 9: The two tracks and the shared evaluation stage. The classical track separates localization from classification. YOLO26n performs both jointly. Both produce labelled boxes that are matched against the annotations by the same procedure, which is what makes the comparison in Section 6.5 meaningful.

4.2 Shared preprocessing and constants

Both tracks use RGB images and the split in Table 1. The fixed constants in Table 5 ensure that configuration comparisons vary one factor at a time.

Constant	Value	Purpose
CROP_SIZE	128×128	Fixed square input of the classical classifier
PROPOSAL_LONG_SIDE	640 px	Working resolution of the proposal stage
YOLO_IMG_SIZE	640 px	Detector input size
TOP_K_PROPOSALS	2	Proposals retained per image after area ranking
IOU_MATCH	0.5	IoU threshold of the matching protocol
RANDOM_STATE	42	Split, sampling, initialization, and training seed

Table 5: Constants shared by both tracks.

4.3 Classical track

4.3.1 Vegetation mask

The proposal stage assumes that plant tissue is distinguishable by colour. For RGB values R, G, B , Excess Green is

$$\text{ExG}(R, G, B) = 2G - R - B. \quad (1)$$

The index is rescaled to $[0, 255]$ and thresholded by Otsu's method. To reject greenish shadows and soil, it is intersected with OpenCV HSV limits $25 \leq H \leq 95$, $S \geq 25$, and $V \geq 25$:

$$M_{\text{green}} = \text{Otsu}(\text{ExG}) \wedge [25 \leq H \leq 95 \wedge S \geq 25 \wedge V \geq 25], \quad (2)$$

The mask receives one 5×5 elliptical opening and two closings to remove isolated pixels and bridge small gaps.

4.3.2 Guarded colour expansion

Because diseased tissue may be yellow or brown, an expanded mask also considers yellow ($15 \leq H \leq 35$, $S, V \geq 40$) and brown ($5 \leq H \leq 25$, $S \geq 40$, $20 \leq V \leq 210$). Discoloured pixels are retained when adjacent to an 11×11 dilation of the green mask, or when a standalone component has area 0.2–50%, aspect ratio at most 10, solidity at least 0.35, and at most one border contact. The resulting expanded mask is denoted M_{exp} . One 3×3 opening and two 5×5 closings complete the mask; the guards limit confusion with soil and litter.

4.3.3 Region proposers

Three mechanisms convert a mask into boxes, and each is evaluated with both M_{green} and M_{exp} , giving six configurations.

Excess Green and HSV boxes external connected components directly. **Watershed** separates touching components using distance-transform seeds above 0.35 of the maximum and a twice-dilated 3×3 background. **GrabCut** runs three graph-cut iterations from the mask bounding box padded by four pixels, or from a 5%-inset frame when the mask is empty.

For every proposer, boxes below 0.2% image area or above aspect ratio 10 are discarded; the remainder are clipped and the two largest are retained to suppress fragments.

4.3.4 Handcrafted features

Each crop is resized to 128×128 . The 1,812-value **basic** vector combines a 48-value HSV histogram with a 1,764-value HOG descriptor using 9 orientations, 16×16 cells, 2×2 blocks, and L2-Hys normalization. The 1,906-value **enhanced** vector adds a 48-value LAB histogram, two uniform LBP histograms (8 and 16 neighbours), LAB/HSV moments, and green, yellow, and brown pixel ratios. Histograms are normalized independently and all vectors are checked for finite values.

4.3.5 Classifiers

The linear control is multinomial logistic regression after a training-only **StandardScaler**, with at most 3,000 iterations. Histogram-based gradient boosting uses unscaled features, 100 iterations, at least 20 samples per leaf, 63 bins, and no early stopping. Its eight candidates cross learning rates $\{0.05, 0.10\}$, leaf counts $\{15, 31\}$, and L_2 strengths $\{0, 1\}$.

Both families address imbalance through class weighting rather than resampling. With N training boxes, K classes, and n_c boxes in class c , the weight applied to every sample of class c in the loss is

$$w_c = \frac{N}{K n_c}, \quad (3)$$

which equalizes total class weight. Candidates are selected by validation macro- F_1 , with median recall among classes below 100 training boxes as the tie-breaker.

4.4 Deep track

4.4.1 Model and training configuration

The deep track fine-tunes COCO-pretrained **YOLO26n** [3], a 2.51-million-parameter detector requiring 5.3 GFLOPs at 640×640 . Its compact size and pretrained features suit the many low-support classes; Table 6 gives the configuration.

Item	Setting
Weights	yolo26n.pt, COCO-pretrained, with 2.51 M parameters and 2.38 M after fusing for inference
Input	640 × 640 letterboxed
Schedule	50 epochs, 182 iterations per epoch, early-stopping patience 15, automatic batch size, seed 42, deterministic
Geometric augmentation	Horizontal flip $p = 0.50$, vertical flip $p = 0.25$, rotation $\pm 15^\circ$, translation $\pm 10\%$, and scale 0.75–1.25
Photometric augmentation	HSV jitter $(h, s, v) = (0.01, 0.25, 0.20)$, with brightness and contrast $\pm 15\%$ at $p = 0.25$
Composition	Mosaic $p = 0.50$ and MixUp $p = 0.05$, both disabled for the final 10 epochs
Copy-paste	Disabled: the annotations are boxes, and pasting a rectangle would import background
Evaluation data	The unmodified validation and test directories of Table 1

Table 6: YOLO26n training configuration. A single policy, augmentation combined with class-aware oversampling, is trained.

4.4.2 Class-aware oversampling

Rare classes are visited more often by repeating image paths. Each class receives a multiplier from its training count n_c :

$$m(n_c) = \begin{cases} 4, & n_c < 10, \\ 3, & 10 \leq n_c < 20, \\ 2, & 20 \leq n_c < 100, \\ 1, & n_c \geq 100, \end{cases} \quad (4)$$

Let C_i be the set of annotated classes in training image i . Each image inherits the repetition multiplier $m_i = \max_{c \in C_i} m(n_c)$ so a rare class in a mixed scene is not under-sampled. Capped tiers avoid the extreme duplication caused by inverse-frequency sampling.

The resulting oversampling has **3,984 entries** including 3,342 images occur once, 180 twice, 66 three times, and 21 four times. Counts use training data only, and an audit excludes validation and test paths.

5 Evaluation Metrics

The metrics separate geometry, localization, classification, and confidence-ranked detection so that failures at one stage are not attributed to another.

5.1 Notation

Let $K = 27$ be the number of classes and c a class index. An image with pixel width W and height H has annotations $\mathcal{G} = \{g_1, \dots, g_n\}$ and outputs $\mathcal{P} = \{p_1, \dots, p_m\}$. A box is $b = (x_1, y_1, x_2, y_2)$ with area $|b|$; the matching threshold is $\tau = 0.5$ unless stated otherwise. Here n

and m count annotations and outputs, respectively. Partition-level notation aggregates these sets across images; matching and overlap are computed within each image.

5.2 Box representation and geometry

Normalized annotations $(c_x, c_y, w, h) \in [0, 1]^4$, with centre (c_x, c_y) , width w , and height h , are converted to pixel corners by

$$(x_1, y_1, x_2, y_2) = \left((c_x - \frac{w}{2})W, (c_y - \frac{h}{2})H, (c_x + \frac{w}{2})W, (c_y + \frac{h}{2})H \right), \quad (5)$$

Coordinates are clipped to the image bounds before cropping or overlap calculations.

Three derived quantities describe the shape of the annotations. The relative area, the physical aspect ratio, and the nominal short side at the detector's working resolution are

$$a = w h, \quad (6)$$

$$\rho = \frac{w W}{h H}, \quad (7)$$

$$s^{(640)} = \min\left(1, \frac{640}{\max(W, H)}\right), \quad \sigma = s^{(640)} \cdot \min(w W, h H). \quad (8)$$

Here a measures frame coverage, ρ restores image dimensions to measure physical aspect ratio, $s^{(640)}$ is the resize scale, and σ estimates the short side after a 640-pixel long-side resize. Section 3.3 reports these quantities.

5.3 Intersection over Union

For boxes u and v , the intersection area is

$$I(u, v) = \max(0, \min(u_{x_2}, v_{x_2}) - \max(u_{x_1}, v_{x_1})) \cdot \max(0, \min(u_{y_2}, v_{y_2}) - \max(u_{y_1}, v_{y_1})), \quad (9)$$

and Intersection over Union is

$$\text{IoU}(u, v) = \frac{I(u, v)}{|u| + |v| - I(u, v)} = \frac{|u \cap v|}{|u \cup v|} \in [0, 1], \quad (10)$$

with $\text{IoU} = 0$ for an empty union. IoU is scale-invariant and penalizes both omitted and excess area.

5.4 Localization quality

5.4.1 Matching produced boxes to annotations

Matching is greedy and one-to-one. For each image, initialize the set of assigned annotation indices $U = \emptyset$ and the matched-pair set $\mathcal{M} = \emptyset$. In output order, each p_i is assigned to the unmatched annotation with greatest IoU when that overlap reaches τ :

$$j^*(i) = \arg \max_{j \notin U} \text{IoU}(p_i, g_j), \quad \text{accepted iff } \text{IoU}(p_i, g_{j^*(i)}) \geq \tau, \quad (11)$$

Accepted annotation indices join U and their pairs join $\mathcal{M}$. If no annotation remains available, the output is unmatched. Unmatched annotations are localization false negatives. Both tracks use this rule.

5.4.2 Hit rate and mean best IoU

Hit rate is the fraction of annotations matched under Equation (11):

$$\text{HR}_\tau = \frac{|\mathcal{M}|}{|\mathcal{G}|}, \quad (12)$$

Mean best IoU averages each annotation's largest overlap:

$$\overline{\text{IoU}} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \max_{p \in \mathcal{P}} \text{IoU}(g, p), \quad (13)$$

where an image with no outputs contributes zero. Hit rate is thresholded and one-to-one; mean best IoU is continuous but can credit one output against several annotations. Hit rate is the selection criterion and mean best IoU the tie-breaker.

5.4.3 The retention ceiling

For n_i annotated objects in image i , keeping at most k boxes per image gives the count-only hit-rate ceiling

$$\text{HR}_k^{\max} = \frac{\sum_i \min(n_i, k)}{\sum_i n_i}. \quad (14)$$

The sums run over all images in the partition, assuming perfect placement of retained boxes.

5.5 Per-class classification quantities

On proposal crops, classification uses matched pairs only; on ground-truth crops, each crop is paired directly with its annotation. Let $\hat{y}(p)$ be the predicted class and $y(g)$ the annotated class. For class c ,

$$\text{TP}_c = |\{(p, g) \in \mathcal{M} : \hat{y}(p) = c \wedge y(g) = c\}|, \quad (15)$$

$$\text{FP}_c = |\{(p, g) \in \mathcal{M} : \hat{y}(p) = c \wedge y(g) \neq c\}|, \quad (16)$$

$$\text{FN}_c = |\{(p, g) \in \mathcal{M} : \hat{y}(p) \neq c \wedge y(g) = c\}|. \quad (17)$$

From these follow precision, recall, and their harmonic mean,

$$P_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad R_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad F1_c = \frac{2 P_c R_c}{P_c + R_c} = \frac{2 \text{TP}_c}{2 \text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad (18)$$

and undefined ratios are set to zero. Precision measures correctness among predictions, recall measures recovery of true class members, and F_1 is their harmonic mean. Because these counts exclude unmatched annotations, localization failures are reported separately through hit rate and unmatched counts.

5.6 Aggregation across classes

For $N = \sum_c (\text{TP}_c + \text{FN}_c)$ matched objects and $n_c = \text{TP}_c + \text{FN}_c$ evaluated objects of class c , define the following summaries. Here N and n_c refer to evaluation counts, unlike the training counts used in class weighting and oversampling; n_c^{train} explicitly denotes the training box count.

$$\text{Accuracy} = \frac{1}{N} \sum_{c=1}^K \text{TP}_c, \quad (19)$$

$$F1_{\text{macro}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} F1_c, \quad (20)$$

$$F1_{\text{weighted}} = \sum_{c=1}^K \frac{n_c}{N} F1_c, \quad (21)$$

$$R_{\text{rare}} = \text{median}\{R_c : c \in \mathcal{C}, n_c^{\text{train}} < 100\}. \quad (22)$$

For each classification evaluation, $\mathcal{C}$ is the union of class labels observed in its true and predicted labels, matching the notebook's macro averaging without an explicit label list. On ground-truth validation crops this comprises 25 classes, yielding macro- $F_1 = 0.432$ rather than 0.400 when all 27 classes are explicitly included and the two absent classes are assigned zero. On matched proposal crops, $\mathcal{C}$ is formed from the matched subset separately for each method; it need not contain the same 25 classes. Accuracy and weighted F_1 emphasize common classes, whereas macro- F_1 weights classes equally and is therefore primary. Per-class support and median rare-class recall accompany it.

5.7 Detection average precision

Detection metrics use class-aware, one-to-one matching at each IoU threshold. A correctly classified matched detection is a detection true positive; an unmatched detection is a false positive, and an unmatched annotation is a false negative. A wrong-class detection cannot match an annotation of another class. These detection counts include objects excluded from the matched-only classification evaluation of Equations (15)–(17). At a fixed confidence threshold, detection precision and recall are

$$P_c^{\text{det}} = \frac{\text{TP}_c^{\text{det}}}{\text{TP}_c^{\text{det}} + \text{FP}_c^{\text{det}}}, \quad R_c^{\text{det}} = \frac{\text{TP}_c^{\text{det}}}{\text{TP}_c^{\text{det}} + \text{FN}_c^{\text{det}}}. \quad (23)$$

A detector's confidence threshold produces a precision–recall curve. Here r and $\tilde{r}$ denote recall levels and $P_c(r)$ the detector precision at recall r , including unmatched detections as false positives. Its monotone precision envelope is

$$P_c^{\text{interp}}(r) = \max_{\tilde{r} \geq r} P_c(\tilde{r}), \quad (24)$$

and class average precision is its area:

$$\text{AP}_c^r = \int_0^1 P_c^{\text{interp}}(r) dr. \quad (25)$$

Ultralytics evaluates the integral at 101 recall levels. Averaging over classes and IoU thresholds gives

$$\text{mAP@}\tau = \frac{1}{K'} \sum_{c \in \mathcal{C}_{\text{det}}} \text{AP}_c^\tau, \quad \text{mAP@[0.5:0.95]} = \frac{1}{10} \sum_{\tau \in \{0.50, 0.55, \dots, 0.95\}} \text{mAP@}\tau, \quad (26)$$

where $\mathcal{C}_{\text{det}}$ contains classes with ground-truth support in the evaluated partition and $K' = |\mathcal{C}_{\text{det}}| = 25$ on validation and test. This detection class set is separate from the classification averaging set above. AP evaluates confidence ranking, while the multi-threshold average also rewards boundary precision. Reported detector precision and recall use the F_1 -maximizing threshold.

6 Experiments and Results

All selections below are made on the validation partition of Table 1, comprising 773 photographs and 1,318 annotated objects. The test partition, 774 photographs and 1,265 objects, is evaluated once for the configuration that validation selected. Each table cites the equation that defines its columns.

6.1 Localization performance of the proposal configurations

Table 7 evaluates all six configurations of Section 4.3 on the full validation partition.

Mechanism	Mask	Hit rate $\text{HR}_{0.5}$	Mean best IoU	Boxes / image
Watershed	green	32.9%	0.364	1.94
Watershed	expanded	31.9%	0.365	1.86
Excess Green and HSV	green	27.1%	0.339	1.90
GrabCut	green	26.3%	0.327	1.25
GrabCut	expanded	26.3%	0.327	1.25
Excess Green and HSV	expanded	24.4%	0.316	1.73

Table 7: Proposal quality on the 773 validation photographs and their 1,318 annotated objects. Hit rate follows Equation (12), mean best IoU follows Equation (13), and both use the matching of Equation (11) at $\tau = 0.5$. Rows are ordered by hit rate, which is the selection criterion.

Watershed with the green mask is selected, reaching 32.9% hit rate versus 27.1% for connected components; GrabCut returns only 1.25 boxes per image and performs worst. Colour expansion lowers watershed hit rate to 31.9% even though mean best IoU rises from 0.364 to 0.365, showing why hit rate, rather than average overlap, is the selection criterion. The expansion mostly enlarges existing components with plant tissue or litter instead of recovering objects.

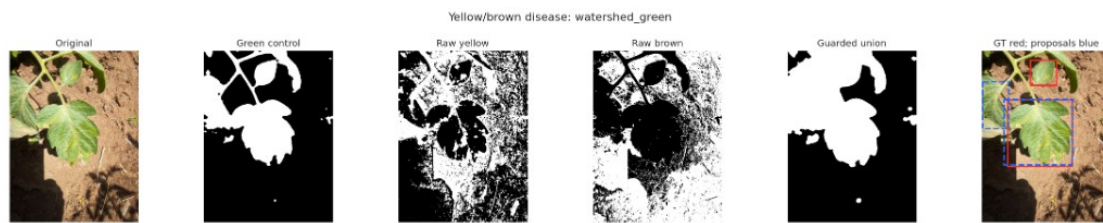


Figure 10: A yellow and brown disease photograph. From left: the green control mask, the raw yellow response, the raw brown response, the guarded union of Section 4.3.2, and the watershed proposals in blue against the red annotation. The raw responses are mostly scattered specks, so the guarded union stays close to the control and does not recover the small annotation.

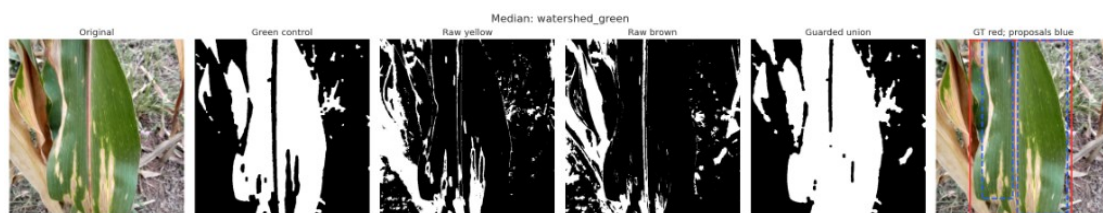


Figure 11: A median-coverage corn leaf. The green control drops the yellow streaks. The guarded union restores those touching green tissue, so the mask follows more of the leaf. The annotation in red still covers the whole leaf while the proposals in blue remain partial strips, which is the pattern that raises mean best IoU without raising the hit rate.

Val: human boxes vs best classical proposer (xyxy, top-2)



Figure 12: Validation examples comparing the annotations in red (top) with the boxes in green (bottom) produced by the selected configuration, watershed on the green control mask.

Figure 12 shows merging, fragmentation, shadow responses, and large background components, explaining the low observed localization coverage.

6.2 Classification performance on ground-truth crops

The classification stage is evaluated first in isolation, on crops taken from the annotations themselves. This isolates classification from localization and provides a reference for the proposal-crop evaluation in Section 6.3. Table 8 compares the three stages.

Stage	Dim.	Macro- F_1	Accuracy	Weighted F_1	Rare recall
Basic features, balanced logistic regression	1,812	0.324	0.589	0.593	0.091
Enhanced features, balanced logistic regression	1,906	0.382	0.662	0.662	0.200
Enhanced features, balanced gradient boosting	1,906	0.432	0.778	0.758	0.091

Table 8: Validation performance on ground-truth crops. Macro- F_1 follows Equation (20), accuracy Equation (19), weighted F_1 Equation (21), and rare-class recall Equation (22). The gradient boosting row is the candidate selected from the grid of Table 9.

Enhanced features raise logistic-regression macro- F_1 from 0.324 to 0.382, and gradient boosting raises it to 0.432 with accuracy 0.778. Enhanced logistic regression has higher rare recall

(0.200 versus 0.091), but macro- F_1 is the primary criterion, so enhanced gradient boosting is selected.

Rate	Leaves	L_2	Macro- F_1	Accuracy	Rare recall	Fit (s)	
0.05	31	1	0.432	0.778	0.091	109.9	selected
0.05	15	0	0.423	0.780	0.000	103.9	
0.10	31	1	0.422	0.775	0.000	113.0	
0.05	15	1	0.413	0.772	0.000	89.9	
0.10	15	1	0.412	0.778	0.000	90.4	
0.05	31	0	0.404	0.772	0.000	158.2	
0.10	31	0	0.382	0.778	0.000	135.3	
0.10	15	0	0.366	0.773	0.000	100.2	

Table 9: The eight gradient boosting candidates on validation ground-truth crops, ordered by macro- F_1 then by rare-class recall. All use 100 boosting iterations, at least 20 samples per leaf, 63 bins, balanced class weights, and no early stopping.

Across Table 9, accuracy varies only from 0.772 to 0.780 while macro- F_1 varies from 0.366 to 0.432. The selected lower-rate, L_2 -regularized model is also the only candidate with non-zero median rare-class recall.

6.3 End-to-end attribution of the classical pipeline

Combining the selected proposer with the selected classifier gives the complete classical system. Table 10 reports it beside a control in which only the classifier is downgraded, so that the effect of the proposal stage can be read off directly.

Configuration	Split	GT-crop	End-to-end	HR _{0.5}	Matched	Unmatched
Watershed green, basic logistic regression	val	0.324	0.258	32.9%	434	884
Watershed green, enhanced boosting	val	0.432	0.348	32.9%	434	884
Watershed green, enhanced boosting	test	0.477	0.387	30.3%	383	882

Table 10: Classical end-to-end attribution. “GT-crop” is macro- F_1 on annotation crops, the reference for the classification stage. “End-to-end” is macro- F_1 on matched proposal crops. Unmatched objects are localization false negatives per Equation (11) and are excluded from the proposal-crop score; the GT-crop score evaluates all annotation crops.

On validation, imprecise proposal crops reduce macro- F_1 from 0.432 to 0.348. The larger loss is localization: only **434 of 1,318 objects** are matched, leaving 884 without any classification. Test macro- F_1 is higher despite a lower 30.3% hit rate.

6.4 Detector results

Metric	Validation	Test
mAP@0.5	0.695	0.744
mAP@[0.5:0.95]	0.548	0.609
Precision	0.686	0.676
Recall	0.689	0.675
Objects	1,318	1,265

Table 11: YOLO26n detection metrics, following Equation (26). Precision and recall are read at the confidence threshold maximizing F_1 and are therefore a single operating point on the curve that mAP integrates.

The validation gap of 0.147 between mAP@0.5 and mAP@[0.5:0.95] reflects stricter box fitting. Precision and recall are nearly equal at the F_1 -maximizing threshold. Test mAP is 0.049 higher at IoU 0.5.

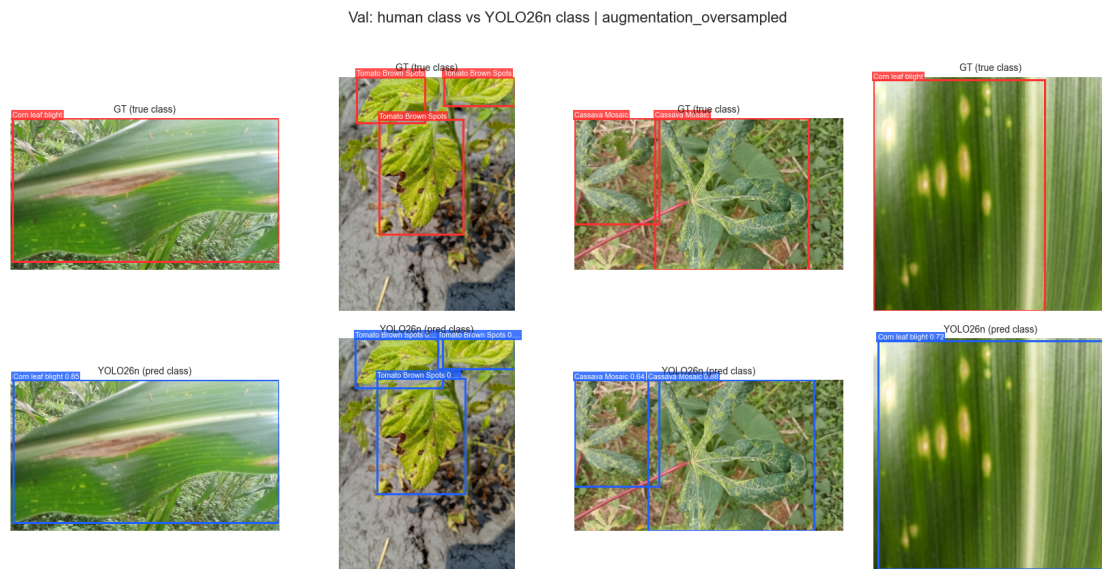


Figure 13: Validation examples with annotations in red and YOLO26n predictions in blue. Numbers beside the predicted labels are confidence scores. Predictions are filtered at confidence 0.25 before the matching analysis of Section 6.5. Retained predictions are matched one-to-one at $\text{IoU} \geq 0.5$; confidence does not weight the matched scores.

Figure 13 shows boxes following object rather than colour boundaries, including discoloured and adjacent leaves. YOLO26n also has no two-output ceiling.

6.5 Two tracks comparison

The two tracks are now compared under one protocol. Both are matched to the same annotations by Equation (11) at $\tau = 0.5$. YOLO26n predictions are first filtered at a fixed confidence threshold of 0.25, as in the notebook's parity evaluation. Classical proposals have no confidence score and use the top-two retention rule. Macro- F_1 is computed on matched pairs only, with the class set defined in Section 5.6; unmatched objects are excluded from this classification score and reported separately through hit rate and unmatched counts. Confidence does not weight the matched score, but its filtering threshold affects which YOLO predictions are available for matching.

Method	Matched macro- F_1	HR _{0.5}	Matched objects	Unmatched objects
Watershed green proposals with enhanced boosting	0.348	32.9%	434	884
YOLO26n with augmentation and oversampling	0.802	84.4%	1,113	205

Table 12: Validation parity between the two tracks. Matched macro- F_1 follows Equation (20) over matched pairs, and hit rate follows Equation (12). Both are computed over the same 1,318 annotated objects.

YOLO26n matches **1,113 of 1,318 objects** versus 434, giving 84.4% versus 32.9% hit rate, and raises matched macro- F_1 from 0.348 to 0.802. The joint comparison shows a substantial increase in both localization coverage and matched classification performance.

6.6 Error analysis

6.6.1 Failure modes of the classical pipeline

On annotation crops, **six of 25 evaluable classes have $F1_c = 0$** ; all have fewer than 20 training boxes and at most five validation objects. High-scoring classes mainly show broad colour or pattern changes captured by global histograms, whereas small localized lesions are diluted. The failures therefore reflect both low support and loss of spatial detail.

Appendix A gives the full per-class report.

6.6.2 Failure modes of the detector

Tomato leaf mosaic virus is worst, with zero recall and $AP@0.5 = 0.091$ over four validation objects. Four classes reach recall 1.000 on at most four objects each; median low-support recall is 0.752. There are 510 unmatched predictions. Tomato healthy shows the largest test regression, plausibly related to its frequent co-occurrence with Tomato blight leaf.

7 Limitations

1. **Dataset coverage and leakage.** The duplicate groups described in Section 3 can inflate held-out performance. The absent classes and sparse support documented in Section 3.1 restrict held-out claims and make rare-class estimates unstable. Class weighting and repetition cannot create missing variation. Differences between validation and test scores on this single split do not establish improved generalization.
2. **Different classification populations.** Proposal-crop macro- F_1 excludes unmatched annotations and predictions and must be read with hit rate and unmatched counts. The two tracks match different subsets, so their classification scores do not measure performance on identical objects. Ground-truth-crop scores are a separate reference, not a strict upper bound on scores from a selected subset. Detection AP also uses a different population and confidence ranking.
3. **Greedy matching and geometric summaries.** Equation (11) may accept fewer pairs than an optimal bipartite assignment. Although shared, this protocol may affect the two methods differently. IoU alone also cannot distinguish displacement, size errors, and merged objects, motivating qualitative error inspection.
4. **Annotation completeness.** Some unmatched predictions may be unlabelled plants rather than true false positives; confirming this requires manual annotation review.
5. **No controlled detector ablations.** The detector uses augmentation and oversampling together. Without separate ablations, their individual contributions cannot be determined.

8 Conclusion and Future Work

8.1 Conclusion

FieldPlant is highly imbalanced, and colour-based localization is the main classical bottleneck. The selected watershed pipeline reaches 32.9% hit rate and matched macro- $F_1 = 0.348$, whereas YOLO26n reaches 84.4% and 0.802 under the same protocol. YOLO26n also obtains validation mAP@0.5 = 0.695. These results favour joint detection for this experiment.

8.2 Future work

Future work should rebuild duplicate-safe partitions, compare a learned class-agnostic proposer with the handcrafted classifier, and run controlled detector ablations for augmentation and oversampling. Additional data collection should target the ten classes with fewer than 20 training boxes.

Appendix A Per-class Results

Table 13 reports validation performance on annotation crops. Zero-support rows identify classes absent from held-out data.

Class	Precision	Recall	$F1_c$	Support
Cassava Bacterial Blight	0.000	0.000	0.000	5
Cassava Brown Leaf Spot	0.556	0.294	0.385	34
Cassava Healthy	0.800	0.686	0.738	70
Cassava Mosaic	0.849	0.944	0.894	285
Cassava Root Rot	0.333	0.091	0.143	11
Corn Brown Spots	0.875	0.226	0.359	31
Corn Charcoal	0.000	0.000	0.000	0
Corn Chlorotic Leaf Spot	0.500	0.250	0.333	4
Corn Gray leaf spot	0.500	0.100	0.167	10
Corn Healthy	0.545	0.240	0.333	25
Corn Insects Damages	0.000	0.000	0.000	3
Corn Mildew	0.714	0.833	0.769	6
Corn Purple Discoloration	0.000	0.000	0.000	1
Corn Smut	0.000	0.000	0.000	2
Corn Streak	0.667	0.571	0.615	35
Corn Stripe	0.840	0.840	0.840	25
Corn Violet Decoloration	0.000	0.000	0.000	1
Corn Yellow Spots	1.000	0.333	0.500	3
Corn Yellowing	0.753	0.773	0.763	75
Corn leaf blight	0.677	0.875	0.763	256
Corn rust leaf	0.667	0.571	0.615	7
Tomato Brown Spots	0.924	0.943	0.934	298
Tomato bacterial wilt	0.000	0.000	0.000	0
Tomato blight leaf	0.597	0.694	0.642	62
Tomato healthy	0.600	0.349	0.441	43
Tomato leaf mosaic virus	0.000	0.000	0.000	4
Tomato leaf yellow virus	0.647	0.500	0.564	22
Accuracy	0.778			1,318
Macro average (27 rows)	0.483	0.375	0.400	1,318
Weighted average	0.764	0.778	0.758	1,318

Table 13: Per-class validation performance of enhanced features with balanced gradient boosting on ground-truth crops, following Equation (18). Macro averages follow Equation (20): over the 25 evaluable classes the macro- F_1 is 0.432, and over all 27 rows it is 0.400.

Table 14 reports both held-out partitions for the 25 evaluable classes.

Class	Validation				Test			
	<i>n</i>	<i>P</i>	<i>R</i>	AP50	<i>n</i>	<i>P</i>	<i>R</i>	AP50
Cassava Bacterial Blight	5	0.281	0.400	0.209	3	0.222	0.333	0.178
Cassava Brown Leaf Spot	34	0.724	0.471	0.649	29	0.730	0.724	0.762
Cassava Healthy	70	0.724	0.743	0.836	57	0.852	0.707	0.820
Cassava Mosaic	285	0.766	0.811	0.855	284	0.791	0.771	0.840
Cassava Root Rot	11	0.516	0.636	0.538	12	0.849	0.472	0.575
Corn Brown Spots	31	0.696	0.516	0.575	33	0.811	0.788	0.775
Corn Chlorotic Leaf Spot	4	0.645	1.000	0.787	4	0.561	0.750	0.808
Corn Gray leaf spot	10	0.484	0.600	0.628	8	0.620	0.500	0.660
Corn Healthy	25	0.506	0.600	0.585	19	0.581	0.579	0.666
Corn Insects Damages	3	0.489	0.667	0.696	2	0.291	0.500	0.662
Corn Mildew	6	0.604	0.833	0.640	6	0.484	0.833	0.749
Corn Purple Discoloration	1	0.754	1.000	0.995	1	0.696	1.000	0.995
Corn Smut	2	0.943	0.500	0.551	2	0.717	0.500	0.495
Corn Streak	35	0.845	0.624	0.778	41	0.848	0.680	0.793
Corn Stripe	25	0.694	0.635	0.688	24	0.841	0.625	0.764
Corn Violet Decoloration	1	0.364	1.000	0.995	1	0.677	1.000	0.995
Corn Yellow Spots	3	0.839	1.000	0.995	2	0.575	0.500	0.828
Corn Yellowing	75	0.884	0.710	0.799	73	0.847	0.671	0.855
Corn leaf blight	256	0.821	0.574	0.755	239	0.810	0.669	0.800
Corn rust leaf	7	0.690	0.857	0.682	9	0.887	0.877	0.918
Tomato Brown Spots	298	0.796	0.851	0.916	302	0.787	0.844	0.903
Tomato blight leaf	62	0.731	0.758	0.782	62	0.878	0.806	0.900
Tomato healthy	43	0.547	0.698	0.604	33	0.328	0.333	0.276
Tomato leaf mosaic virus	4	1.000	0.000	0.091	3	0.514	0.667	0.753
Tomato leaf yellow virus	22	0.805	0.752	0.756	16	0.702	0.750	0.821
All classes	1,318	0.686	0.689	0.695	1,265	0.676	0.675	0.744

Table 14: YOLO26n per-class precision, recall, and AP@0.5 on validation and test, following Equations (23) and (25). These detection metrics include unmatched detections and annotations, unlike matched-only classification. Support is the number of annotated objects of that class in the partition.

Table 15 gives the training-count multiplier and resulting effective count.

Class	Boxes	<i>m</i>	Eff.	Class	Boxes	<i>m</i>	Eff.
Cassava Bacterial Blight	17	3	51	Corn Streak	176	1	176
Cassava Brown Leaf Spot	148	1	148	Corn Stripe	113	1	113
Cassava Healthy	290	1	290	Corn Violet Decoloration	5	4	20
Cassava Mosaic	1,328	1	1,328	Corn Yellow Spots	11	3	33
Cassava Root Rot	54	2	108	Corn Yellowing	347	1	347
Corn Brown Spots	156	1	156	Corn leaf blight	1,157	1	1,157
Corn Charcoal	1	4	4	Corn rust leaf	33	2	66
Corn Chlorotic Leaf Spot	18	3	54	Tomato Brown Spots	1,367	1	1,367
Corn Gray leaf spot	36	2	72	Tomato bacterial wilt	2	4	8
Corn Healthy	108	1	108	Tomato blight leaf	280	1	280
Corn Insects Damages	12	3	36	Tomato healthy	203	1	203
Corn Mildew	31	2	62	Tomato leaf mosaic virus	12	3	36
Corn Purple Discoloration	6	4	24	Tomato leaf yellow virus	77	2	154
Corn Smut	9	4	36				

Table 15: Class-aware repetition tiers from Equation (4) and the resulting effective training box counts. The manifest expands 3,609 image paths to 3,984 entries.

Appendix B Computational Environment

Item	Value
Python	3.12.13
Detection framework	Ultralytics 8.4.163 [3] with PyTorch 2.10.0+cu128
Classical stack	OpenCV, scikit-image, scikit-learn, NumPy, pandas
Accelerator	One NVIDIA Tesla T4, 14,912 MiB

Table 16: Software and hardware environment of the reported run.

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- [3] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, "Ultralytics YOLO26: Unified Real-Time End-to-End Vision Models," arXiv:2606.03748, 2026. The nano detection model used in this report is YOLO26n. <https://arxiv.org/abs/2606.03748>